HEAVY HAUL SEMINAR • APRIL 23-24, 2026

Back Rim Thermal Crack Research

Huseyin Guzel, BNSF Railway
Alicia Lollar, BNSF Railway

BNSF RAILWAY

DALLAS, TX

Back Rim Thermal Crack

Purpose

- Inform the industry on this failure mode
- Share what was learned during BNSF's testing
- Promote further discussion about developing a possible testing method to find these cracks before they reach critical crack size

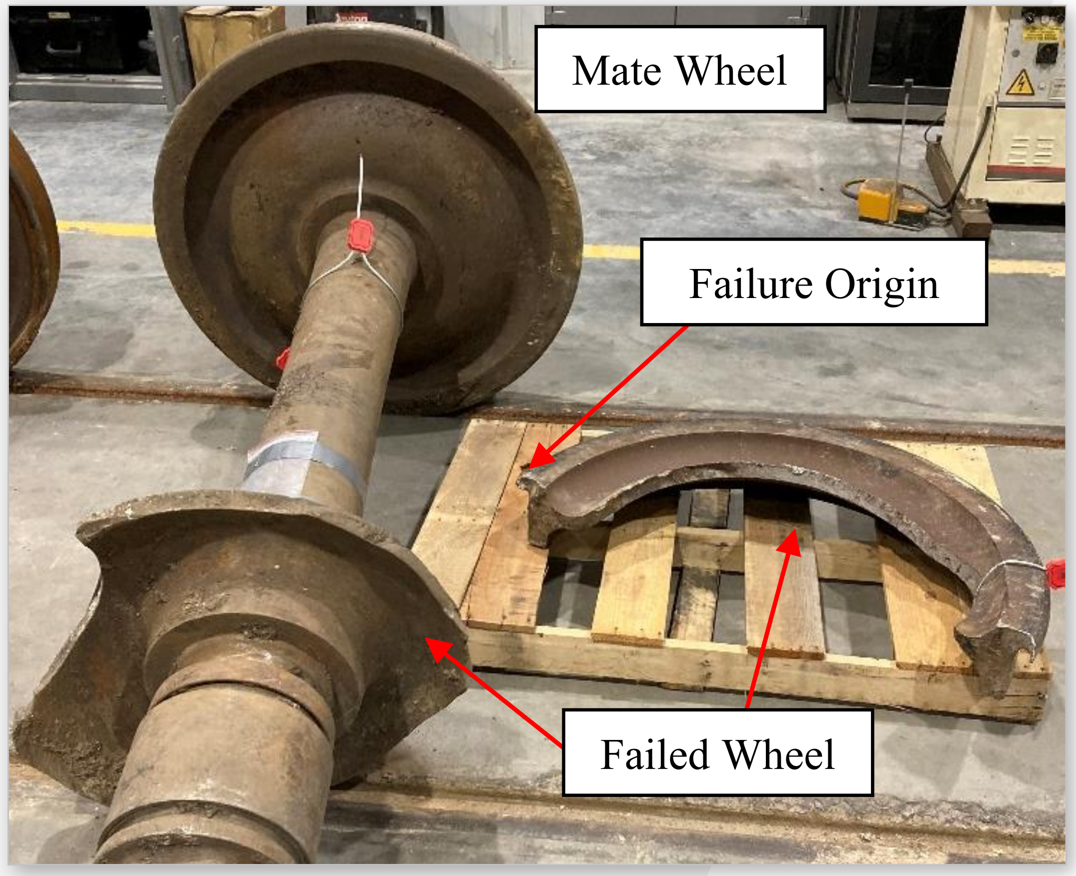
Categorization of Failed Wheels

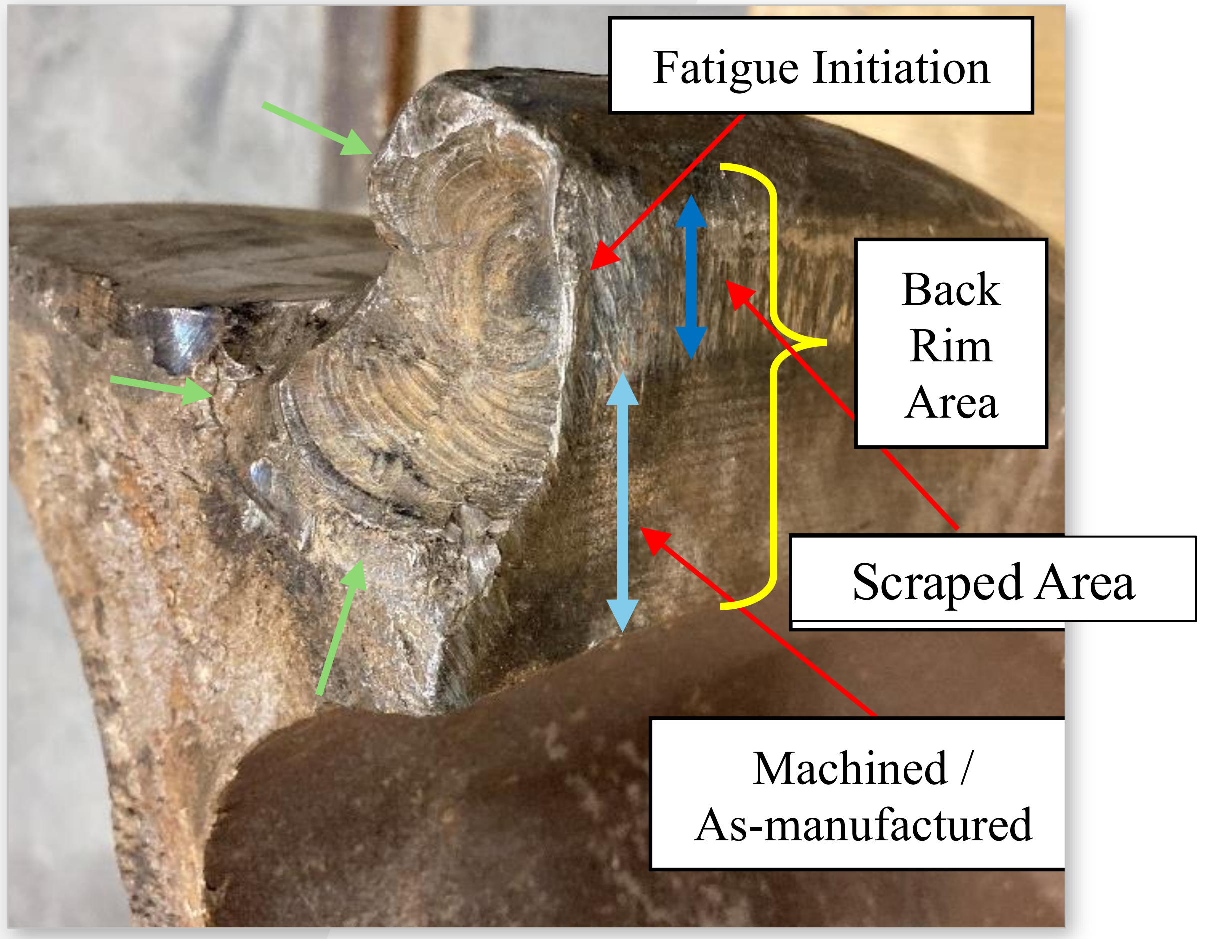
Data was collected for seven of these failures:

- All the failures occurred in either 33 or 38-inch wheels
- Five out of seven cars were intermodal cars. The two most recent failures were on a box and a gondola car
- All the failures originated from scraping on the back rim
- All the wheelsets contained uneven wear between the failed and mate wheels, with the failed wheels showing more wear
- Only one of the wheelsets was reconditioned
- All but one of the failed wheels were on the left side of the car
- Both cast and forged wheels from different manufacturers experienced this type of failure
- Other Class 1 railroads have also experienced this failure

Representative Failure Sample

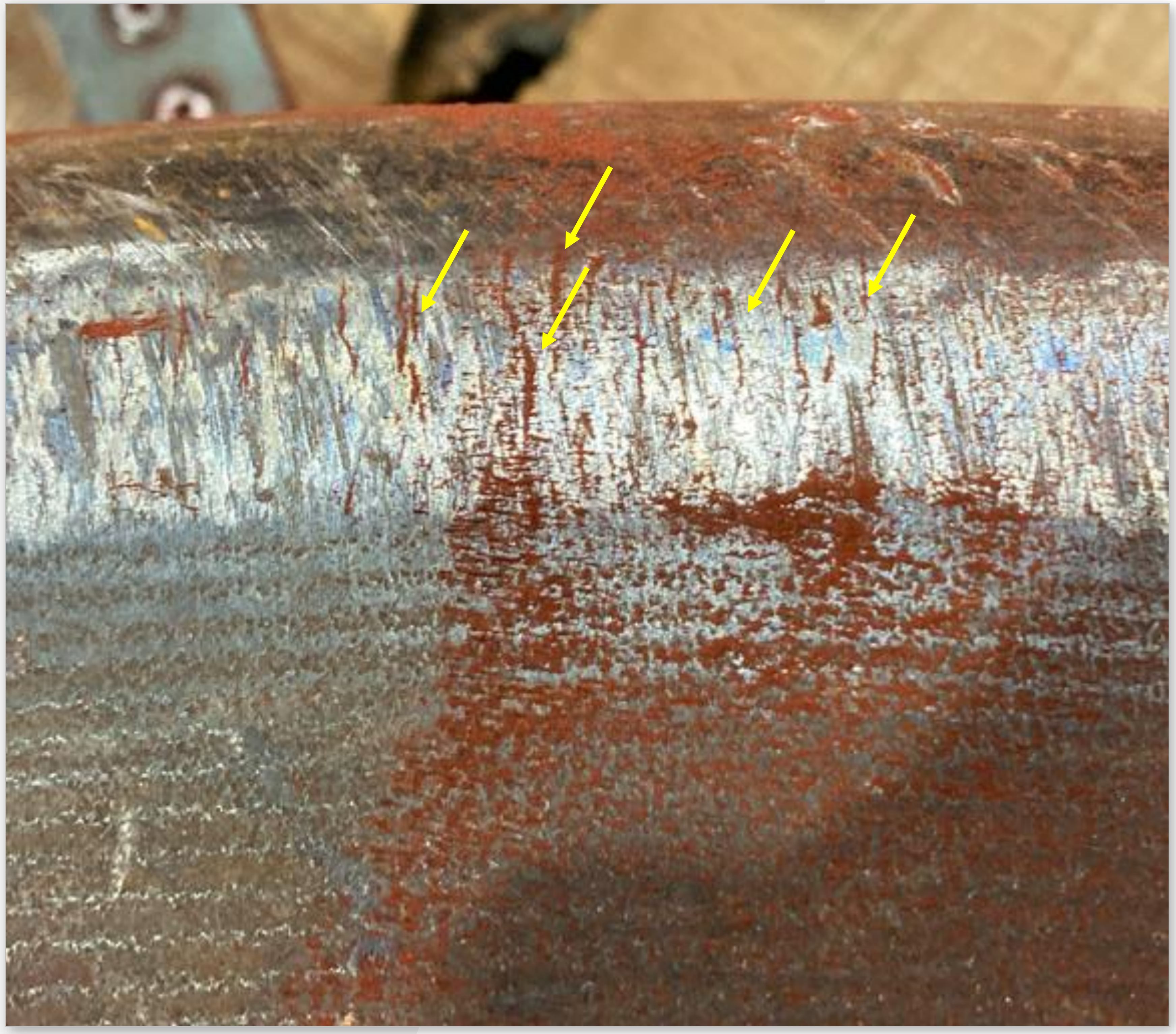
- All the failures have fatigue cracks originating from scraping in the back rim
- On 4/23/23, TR&D inspected the back rim of 40 cars, 544 wheels. All 544 wheels showed signs of scraping on the back rim





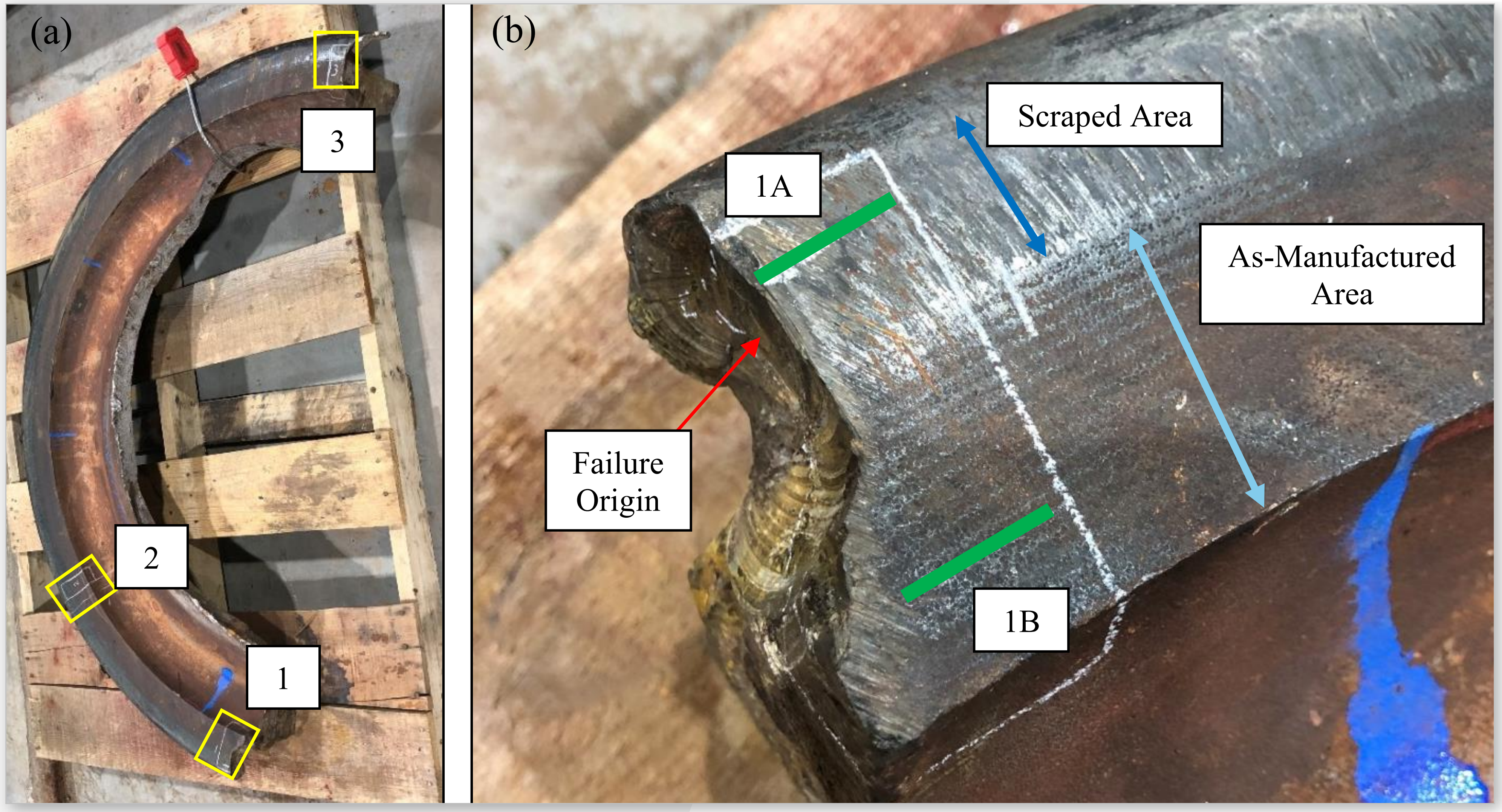
Magnetic Particle Testing

- There were multiple cracks in the scraped area of the back rim surface, as indicated with yellow arrows



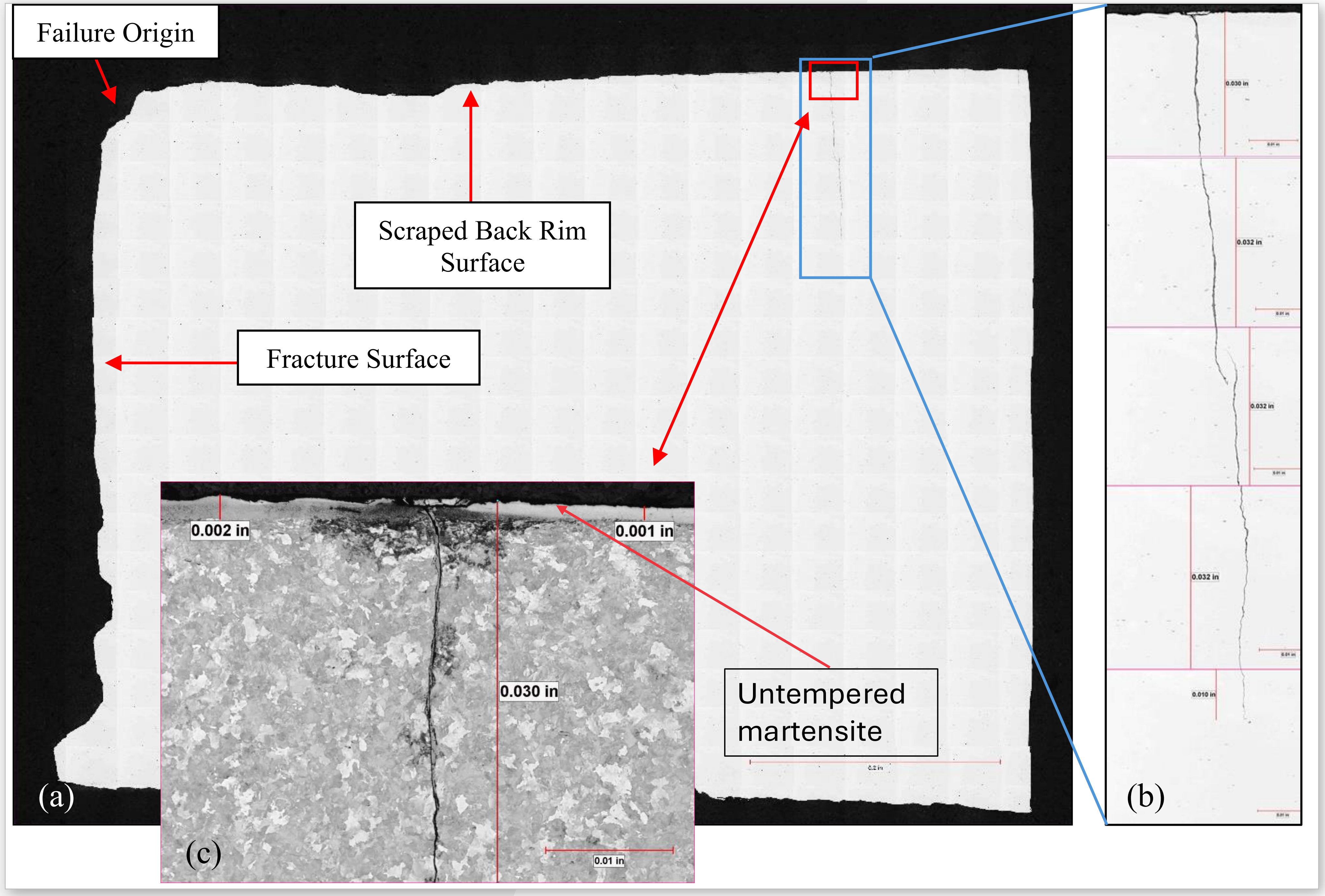
Location of Metallurgical Specimens

- All failed and mate wheels were similarly tested
- Specimens were removed from the scraped areas (1A) and from the as-manufactured areas (1B)



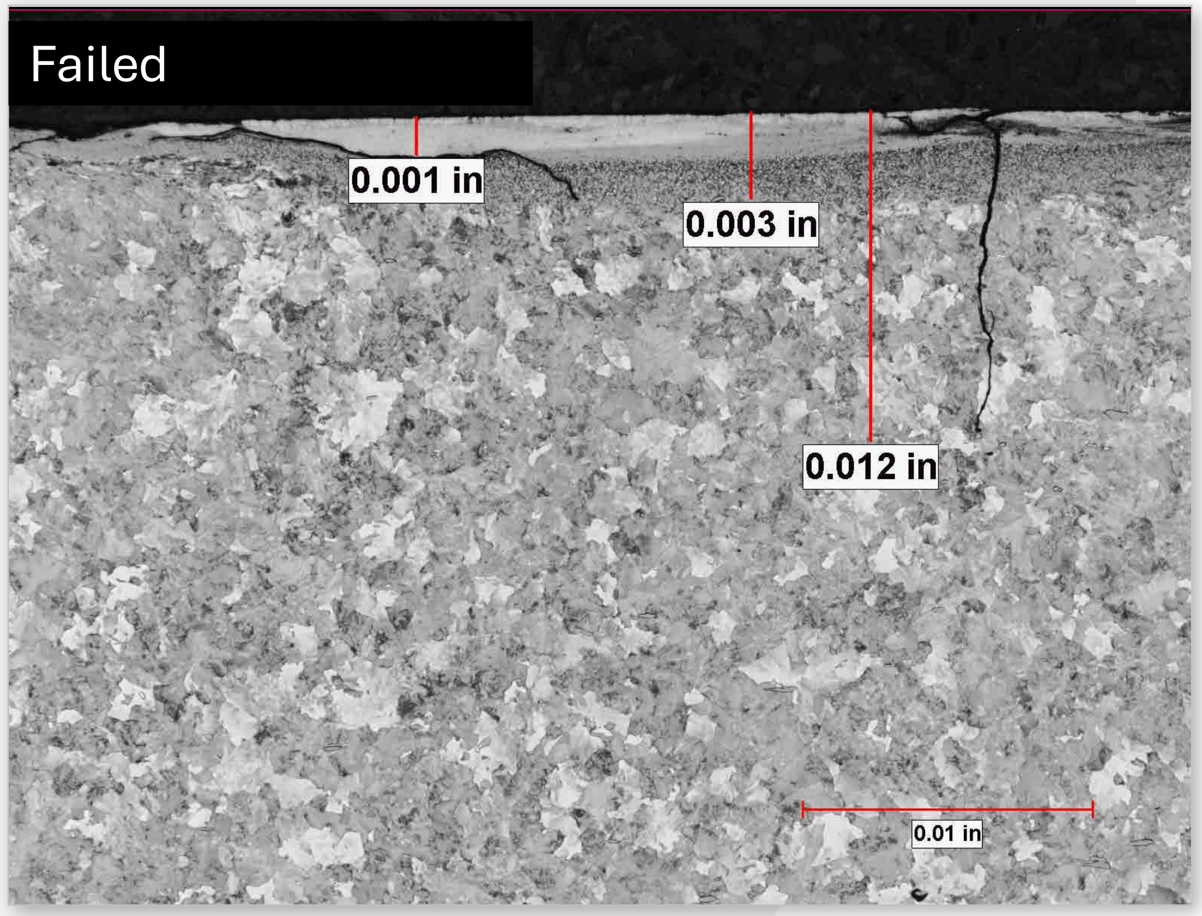
Metallurgical Specimen Example of Failed Wheel

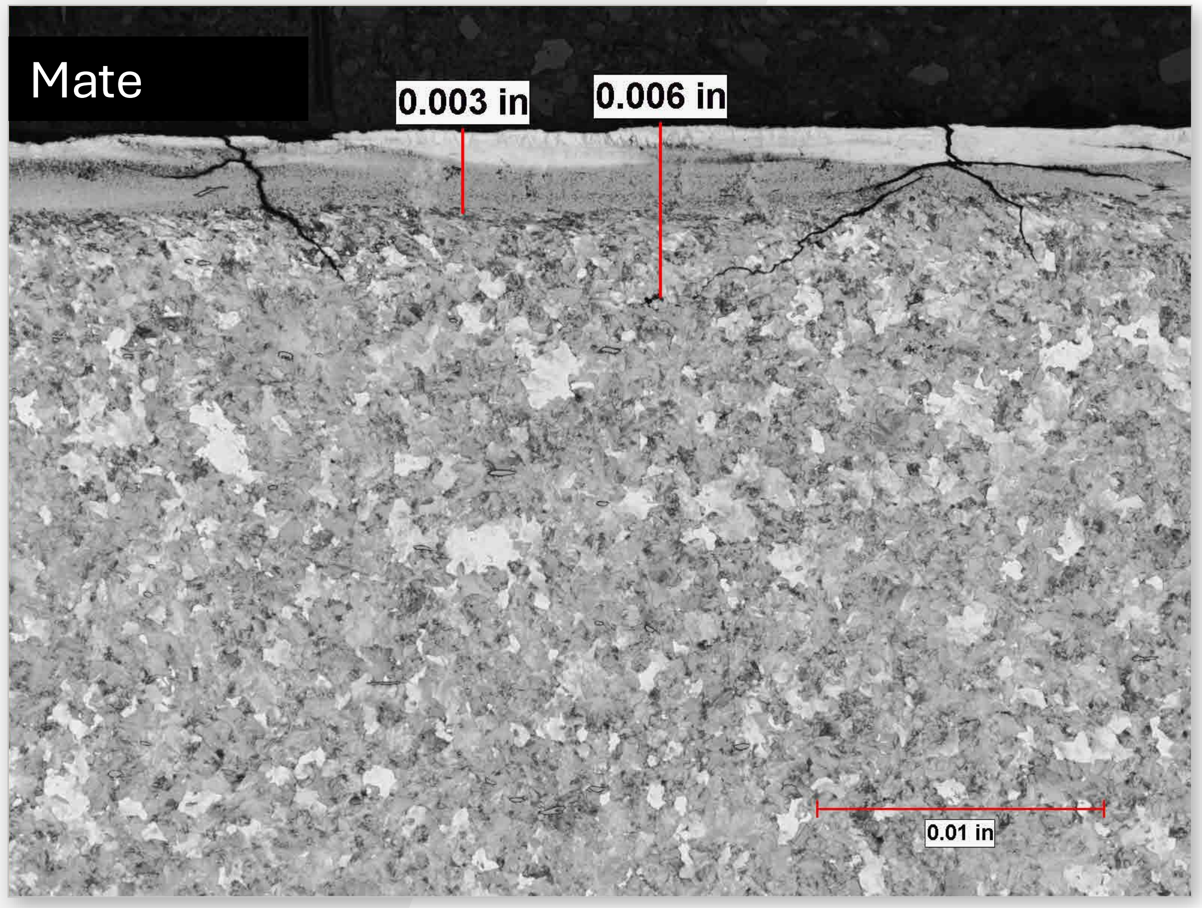
- Multiple cracks grew deeper into the rim of the failed wheel
- The scraped areas contained untempered martensite (white) when etched with Nital



Failed Wheel vs. Mate Wheel (Scraped Area)

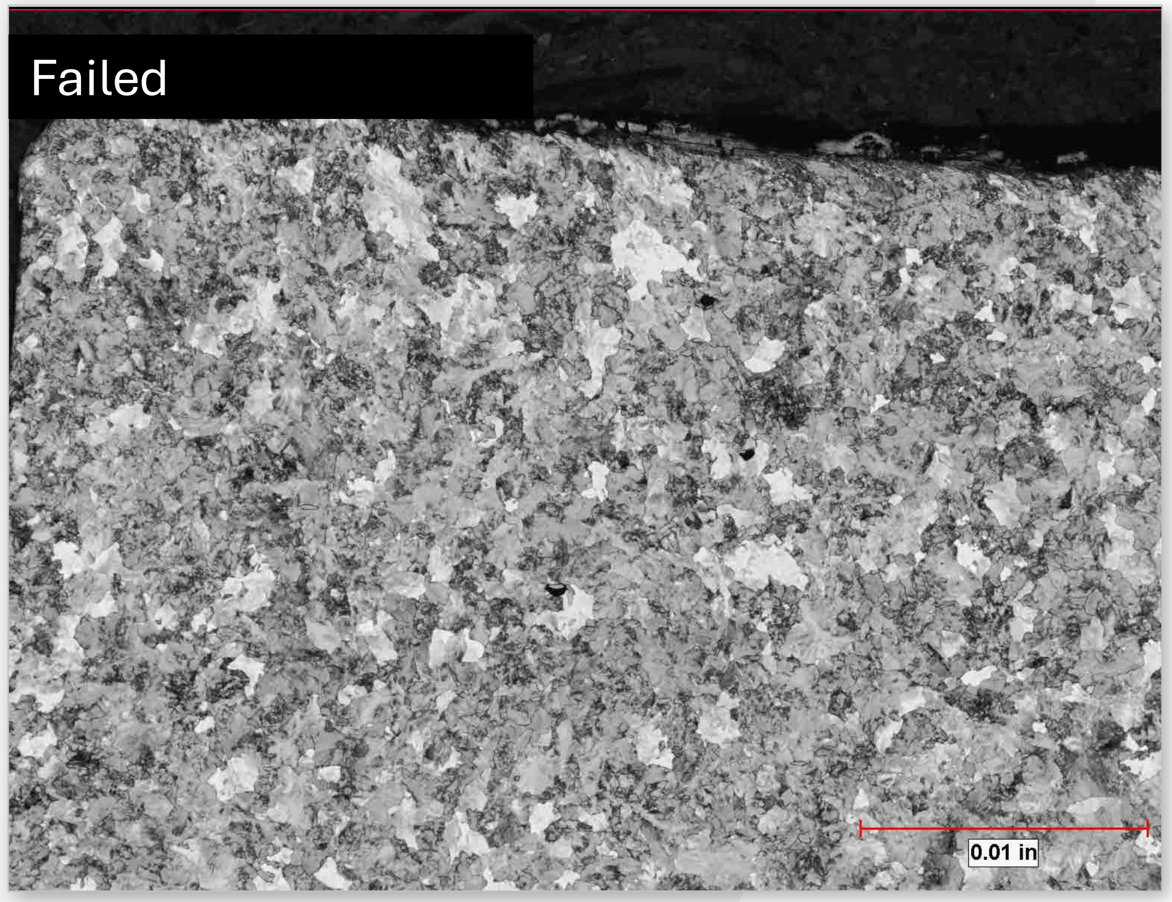
- Microstructures of all the failed and mate wheels examined contained martensite in the scraped area on the back rim surface
- Cracks were found in all the wheels
  - The cracks in the failed wheels (top image) extended deeper into the wheel
  - The cracks in the mate wheels (bottom image) terminated just below the martensitic layer

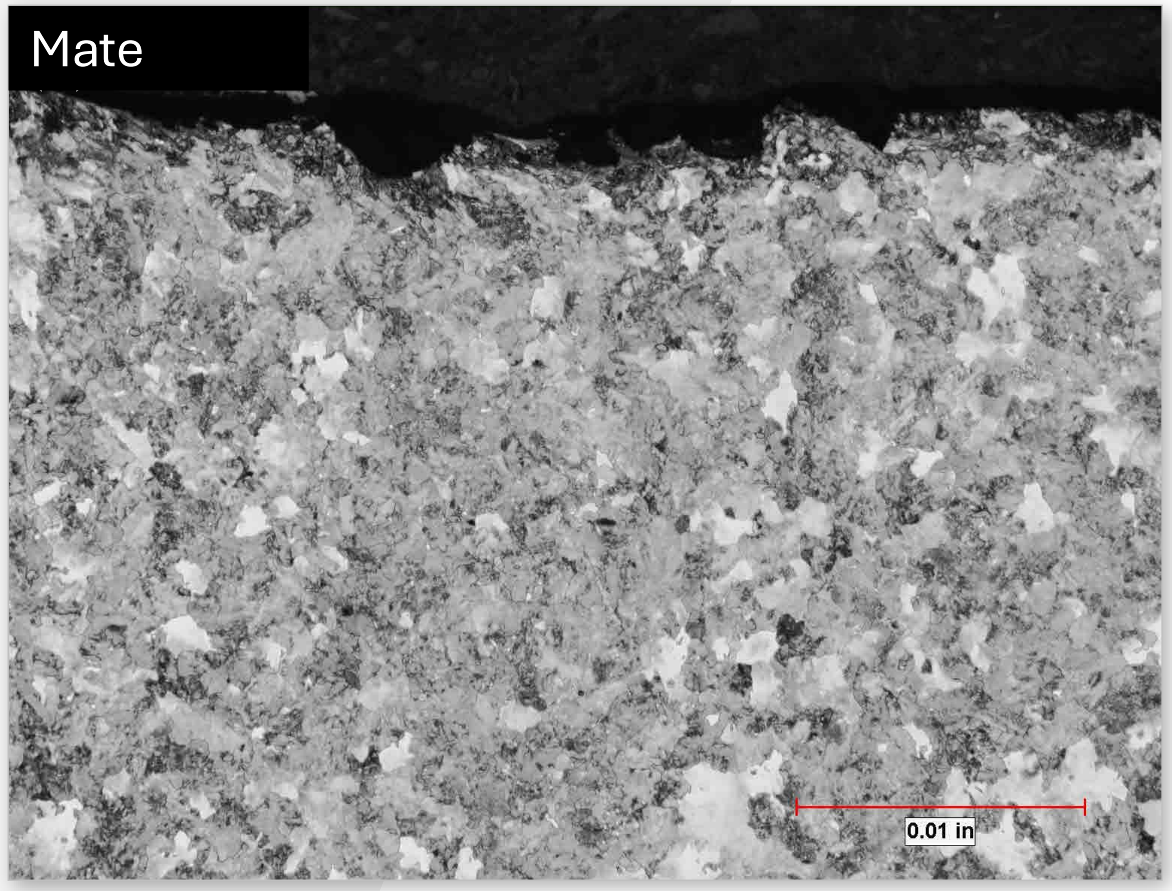




Failed Wheel vs. Mate Wheel (As-Manufactured)

- Metallographic examination of specimens removed from the as-manufactured back rim surface did not show martensite or cracks in either the failed wheels (top image) or the mate wheels (bottom image)





Failed Wheel vs. Mate Wheel – Macroetched Rim Cross Sections

- Macroetch examination of full rim cross sections of the failed wheels (top image) revealed heat damage due to excessive and/or unusual braking, indicating possible issues with the brakes of the rail car
- Macroetch examination of full rim cross sections of the mate wheels (bottom image) did not show heat damage from brakes
- Wheels are heat treated by the manufacturers to induce beneficial compressive hoop stresses in the rim of the wheels. Excessive heat damage may relieve some or all of the compressive hoop stresses

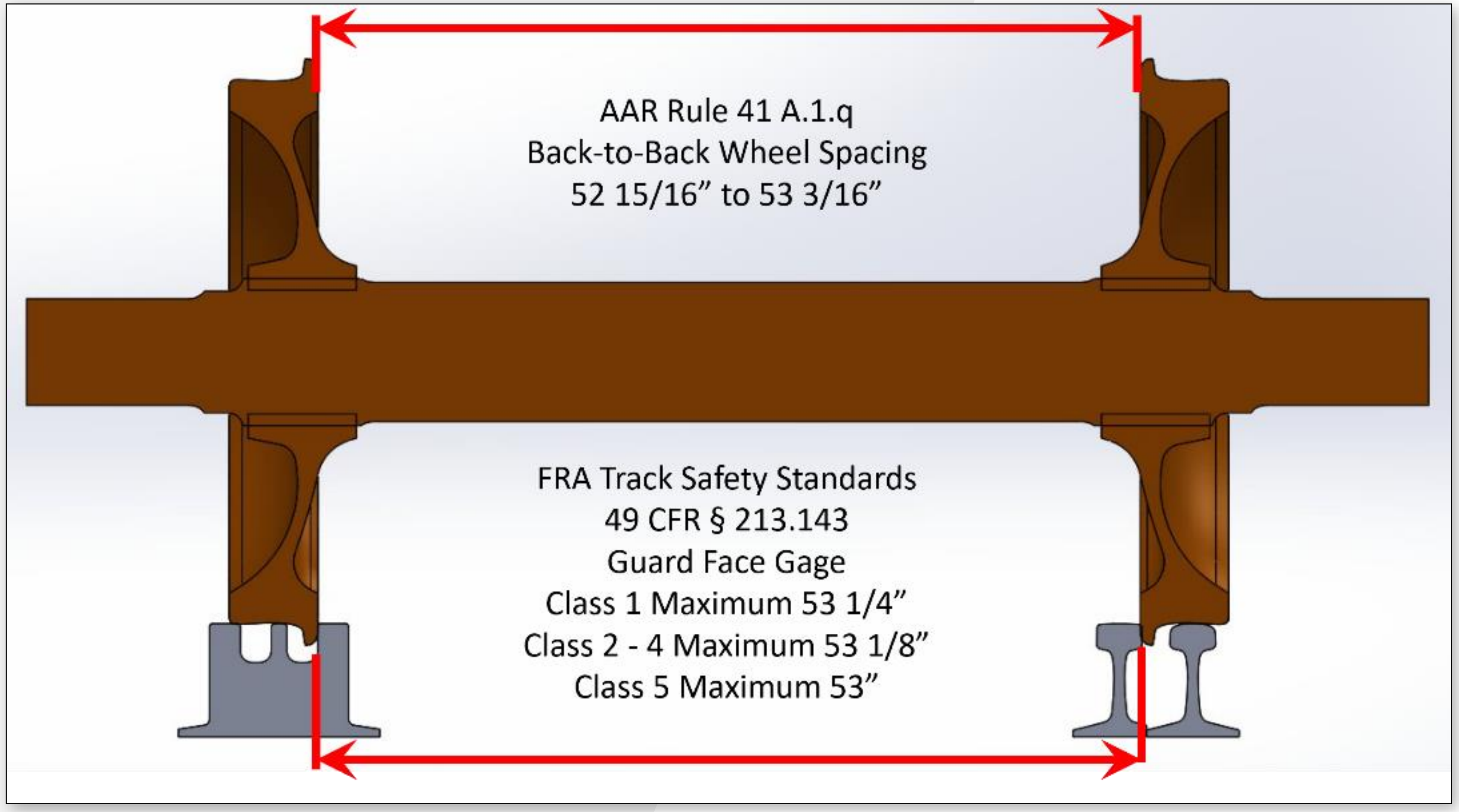
Components Contacting Back Rim

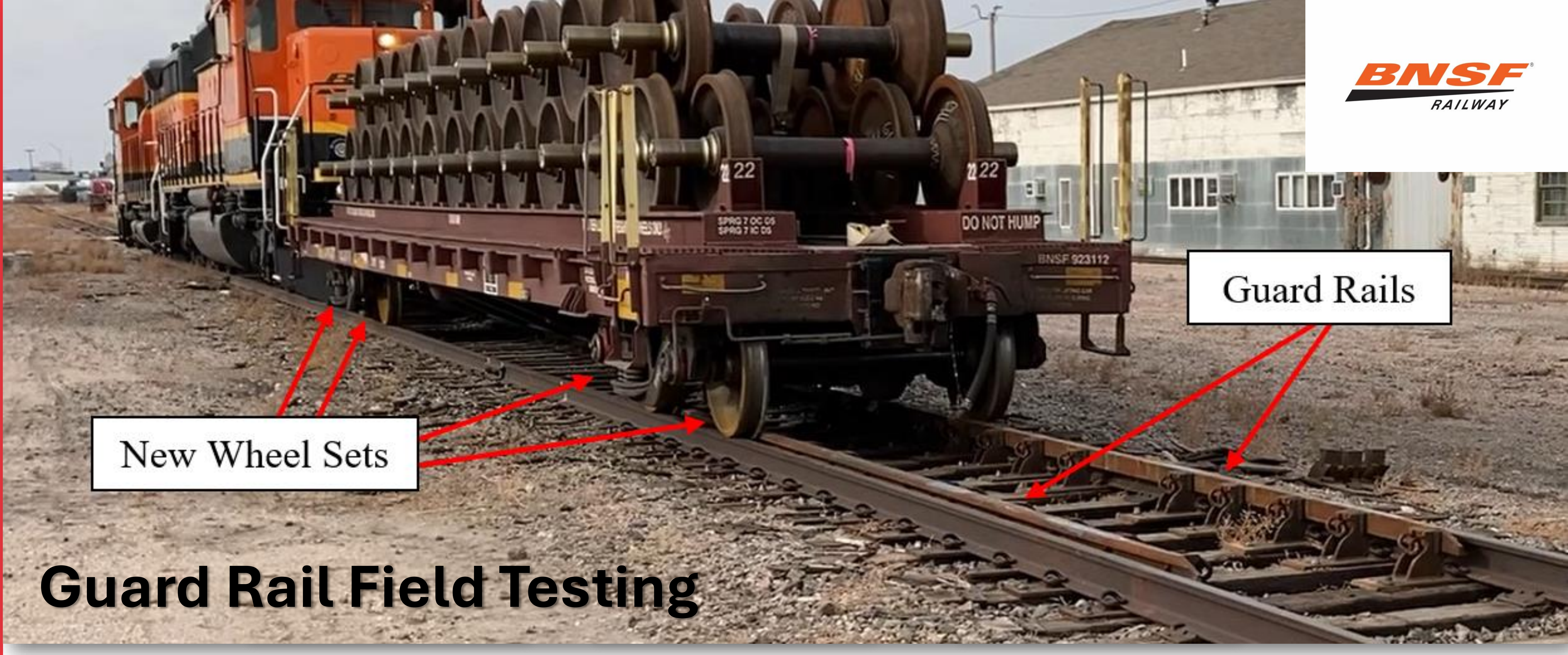
- When wheels travel through guard rails, diamonds, spring frogs, and retarders, the back rim of the wheels make contact with the track components

Guard Rail Interference

- FRA Track Safety Standards allow the guard face gage to be as high as 53-1/4” in Class 1 track
- AAR wheelset standards require the wheel back-to-back spacing to be 52-15/16” to 53-3/16”
- If a guard rail is installed at the maximum allowable guard face gage, every AAR wheelset traversing the guard rail experiences an interference fit

This results in heavy scraping action!



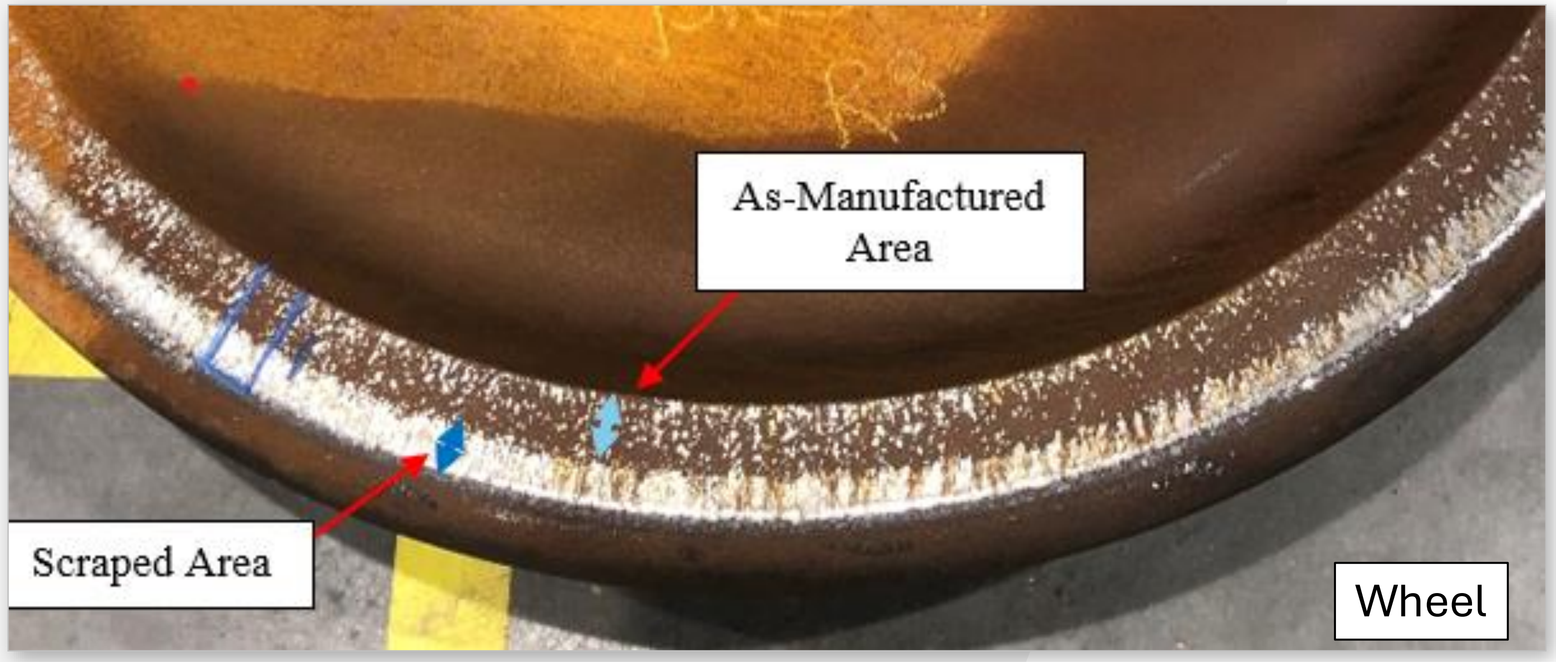


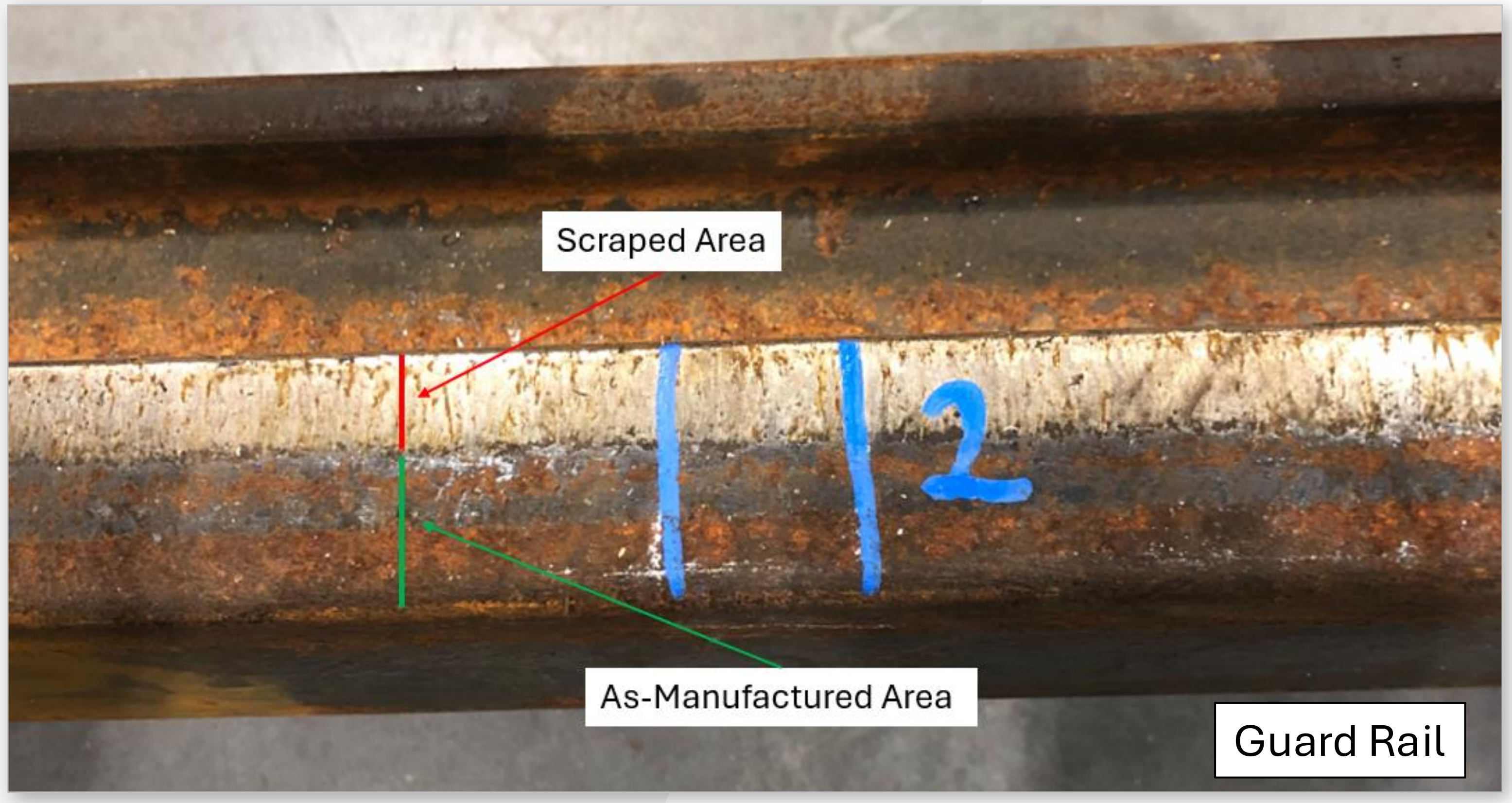
Guard Rail Field Testing

- Field testing was conducted on a yard track north of Havelock yard on November 2, 2022
- The test consisted of two new guard rails installed in tangent track with a guard face gage of 53-1/4” (maximum allowable for Class 1 track)
- New wheels were placed under a loaded car, and the car was pushed/pulled through the guard rails 20 times with the speed ranging from walking speed up to 10 mph

Field Test Wheels and Guard Rails

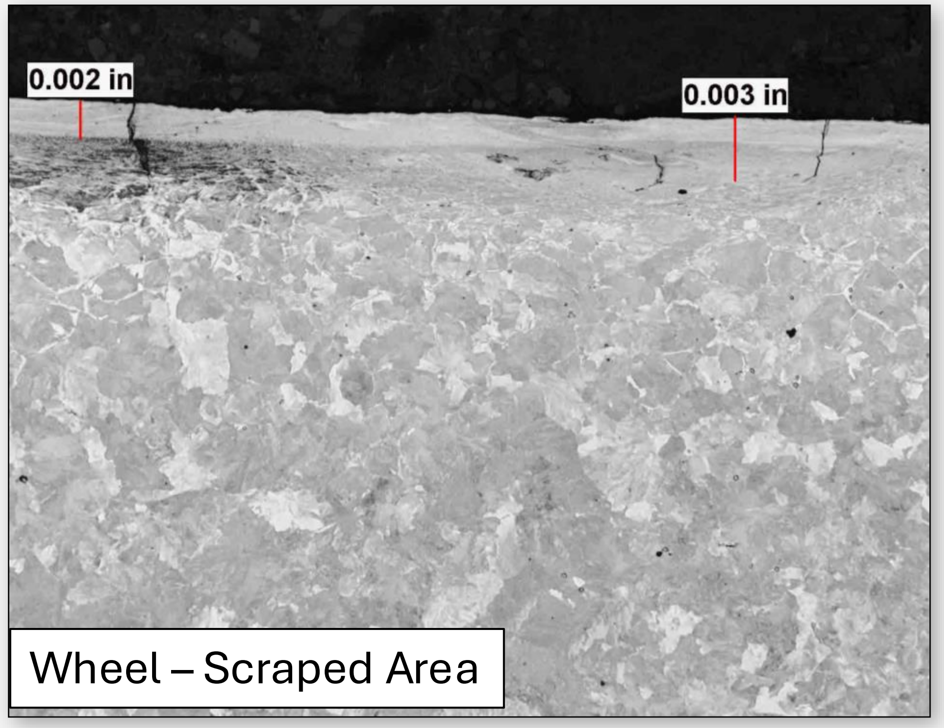
- The wheels and guard rails were removed and received at the lab for metallurgical testing



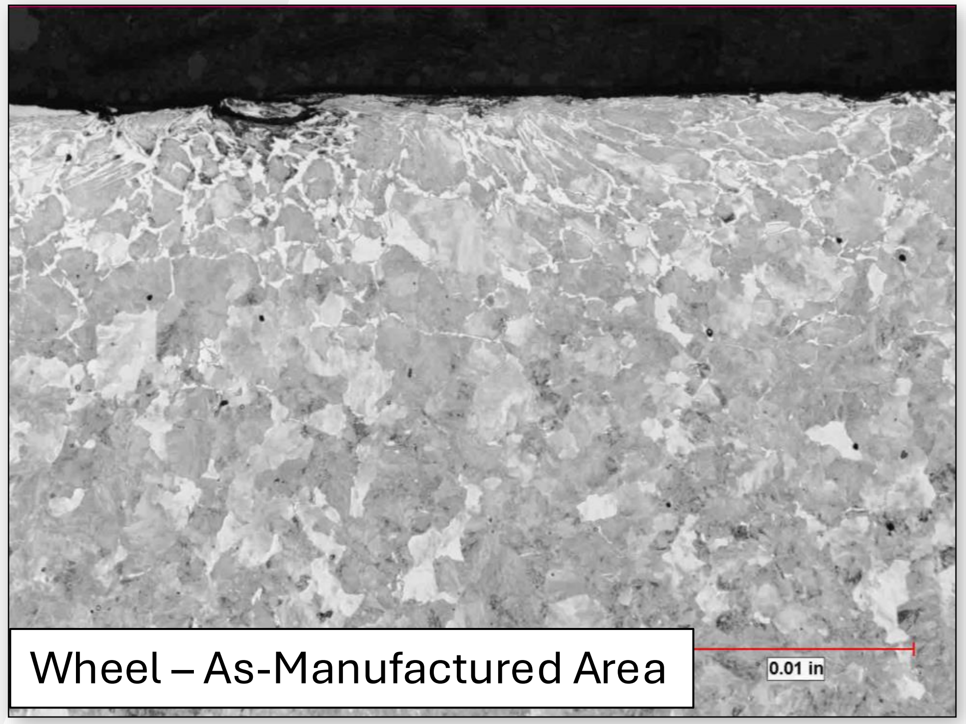


Field Test Metallurgical Testing

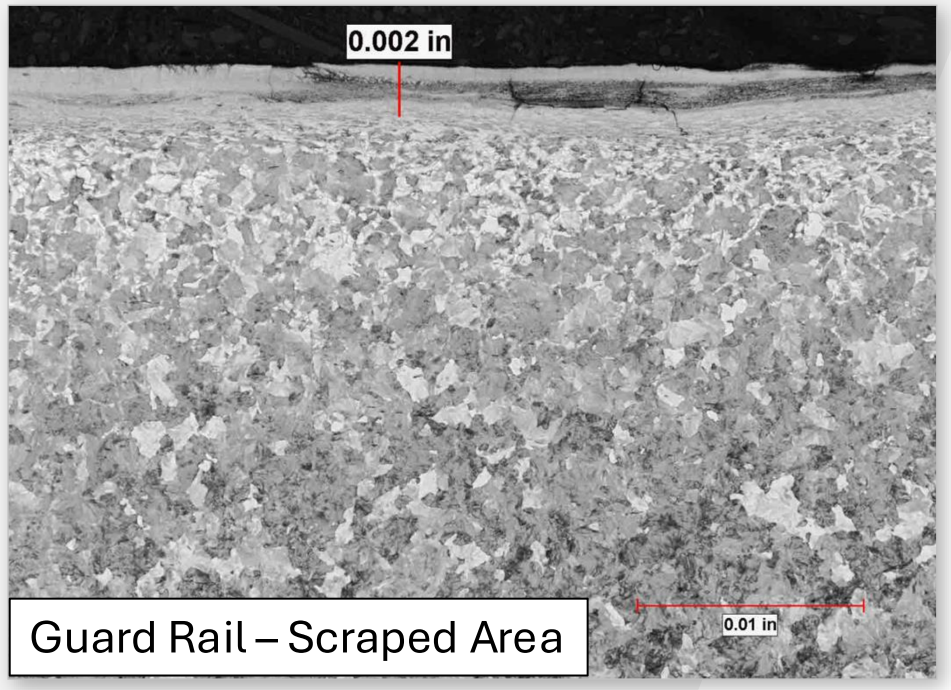
- The specimens removed from the scraped areas revealed cracks and untempered martensite
- The specimens removed from the as-manufactured areas did not show either cracks or martensite, and they contained partial decarburization as expected



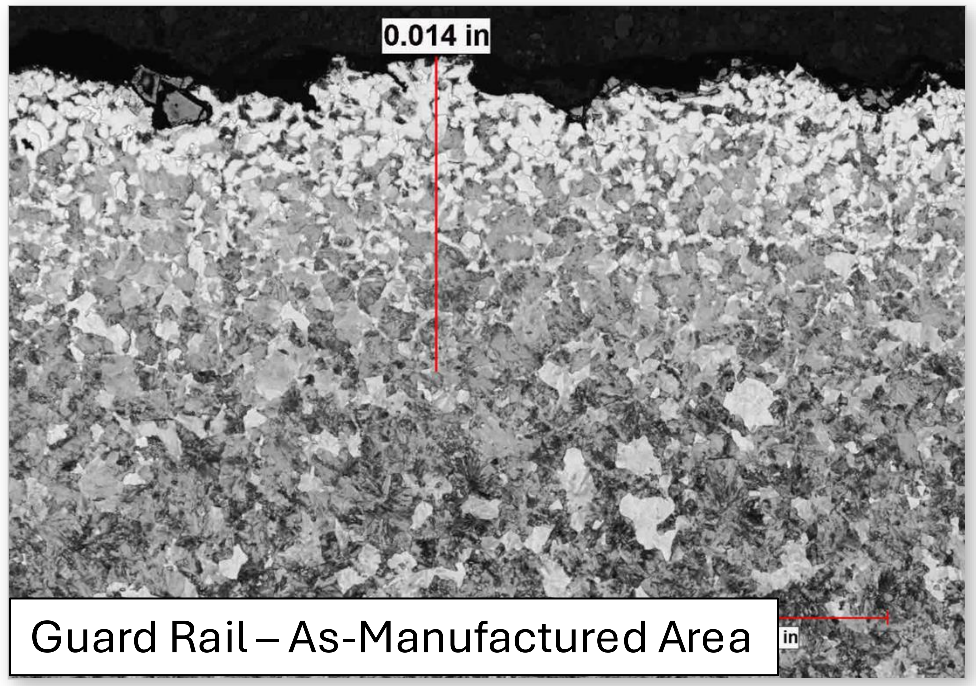
Wheel – Scraped Area



Wheel – As-Manufactured Area



Guard Rail – Scraped Area



Guard Rail – As-Manufactured Area

Findings

- The wheels failed due to thermally assisted fatigue. The fatigue initiated from thermal cracks in the back rim surface where scrape marks were present
- A likely cause of the scrape marks in the back rim is heavy interference contact between this area of the wheel (back rim) and a track component, such as a guard rail
- Metallographic examination of specimens removed from the scraped areas on all wheels, failed and non-failed, revealed numerous cracks which are associated with martensitic transformation
- A significant number of cracks in the back rim of the failed wheels grew deeper into the rim past the martensitic layer
- The cracks in the back rim of the mate wheels were near the surface and generally within the martensitic layer
- Metallographic examination of specimens removed from the as-manufactured back rim surface, with no scraping, did not show martensite or cracks
- Therefore, it is concluded the martensite and associated cracks in the scraped back rim area, which were fatigue crack initiation sites, were created in service due to frictional heating

Findings Continued

- In a field test, interference between guard rails and the back rim surface of the wheel was proven to produce untempered martensite and cracking at the location where these failures originate
- Macroetch examination of a full rim cross section of the failed wheels revealed heat damage due to excessive and/or unusual braking, indicating possible issues with the brakes of the rail cars
- Macroetch examination of a full rim cross section of the mate wheels did not show heat damage from the brakes
- The heat damage due to brakes observed in the failed wheels may have relieved some or all of the beneficial compressive residual hoop stress imparted in the wheel during heat treatment. Loss of the compressive residual hoop stress could be a key factor in allowing fatigue crack growth to progress out of the thermal crack initiation sites generated by heavy scraping in the back rim surface

Recommendations

Based on this study, the following are recommended:

- Industry standards and regulations should be revised to eliminate/reduce the potential for interference between the wheelset back-to-back spacing and guard face gage
- Brake rigging arrangements should be reviewed to eliminate excessive braking
- Further research should be conducted on crack detection methods within the back rim surface of wheels and the mechanisms of their formation and propagation

BNSF RAILWAY

Thank You!

Contact info:

Huseyin.Guzel@bnsf.com
Alicia.Lollar@bnsf.com